

Lecture 21

ChE 467/596-012 Polymer Rheology

Saad A Khan

of

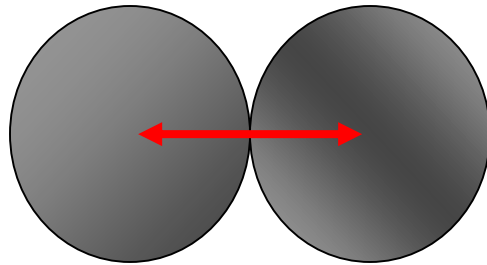
Classification of Dispersion Systems

- “Hard Sphere” systems
 - interparticle interactions screened
- Electrostatically stabilised systems
 - charged particles, electrokinetic phenomena
- Sterically stabilised systems
 - adsorbed polymeric species
 - stability thermodynamic in nature
- Flocculated systems
 - attractive interparticle forces predominate

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“Hard” and “Soft” Particles

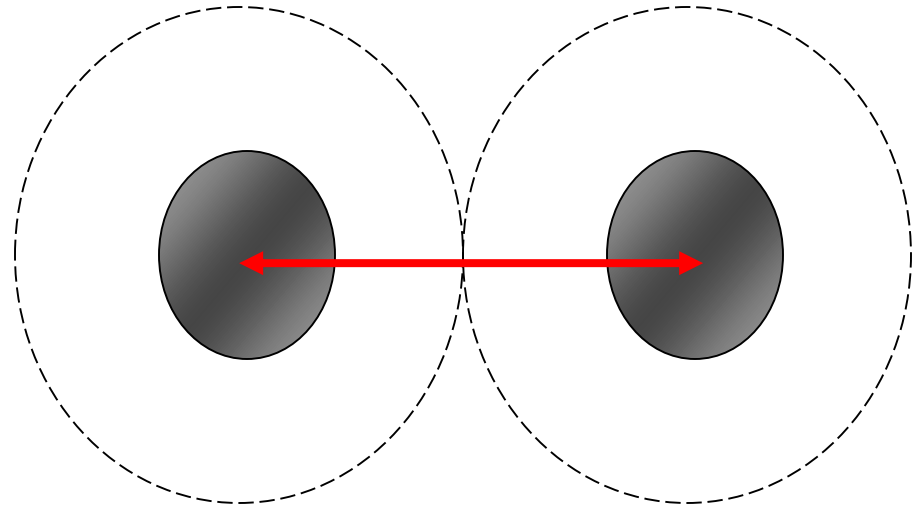
Hard



Colloidal interactions
screened out

Presence of neighbouring
particles experienced only
at close approach
i.e. interparticle separation
 $= 2 \times \text{particle radius}$

Soft



Adsorbed polymer or
electrical double layer

Presence of neighbouring
particles experienced at
greater interparticle separations_{of}

Rheological Behavior

- *3 possible scenarios*
 - *Brownian hard spheres*
 - *Stable suspensions*
 - *Flocculated suspensions*

Brownian hard spheres

- *Neglect dispersion, electrostatic, and steric interactions*
- *Interparticle forces: Brownian, hydrodynamic*
- *Resort to Peclet[#]*
$$Pe^{\#} = \frac{\eta_s \dot{\gamma} a^3}{kT}$$
- *At higher particle concentration, more appropriate to replace η_s with η (suspension)*

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Rheological Behavior

3 possible scenarios

- *Brownian hard spheres*
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Brownian hard spheres

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Brownian hard spheres—2 regimes

- *Dilute regime – already covered most of it*
- *Non-dilute regime*

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Dilute Hard Sphere Systems

Particle volume fraction, $\phi < 0.1$

$$\frac{\eta}{\eta_s} = 1 + 2.5\phi$$

Classical Einstein relationship in which the relative viscosity is a simple linear function of the particle volume fraction

Note - the particle **weight** or **size** does not appear in the above equation.

The factor 2.5 is associated with the intrinsic viscosity of the system, $[\eta]$ -

$$[\eta] = \lim_{\phi \rightarrow 0} \left| \frac{(\eta - \eta_s)}{\eta_s \phi} \right| = \lim_{\phi \rightarrow 0} \left| \frac{\eta_r - 1}{\phi} \right| \text{ of}$$

Dilute Hard Sphere Systems

Effect of particle geometry:

$[\eta] = 2.5$ only holds for spherical particles

Generally, for all particle shapes $\eta_r = 1 + [\eta] \phi$

$[\eta]$ of polymers vs dispersions

Do you see any difference between the $[\eta]$ of polymers and dispersions?

Polymers- $\eta_{\text{rel}} = 1 + [\eta]c + K'[\eta]^2 c^2 + \dots \rightarrow [\eta] = \lim_{c \rightarrow 0} \frac{\eta - \eta_s}{c\eta_s}$

$[\eta]$ of dispersions– no units; for polymers – unit of inverse concentration

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Non-dilute Hard Sphere Systems

Deviation from Einstein equation, in that higher order terms in ϕ are required to describe the viscosity - volume fraction relationship:

$0.1 < \phi < 0.2$ - Bachelor equation, spherical particles:

$$\eta_r = 1 + 2.5\phi + 6.2\phi^2$$

More generally: $\eta_r = 1 + \alpha\phi + \beta\phi^2$

What happens at higher concentration/vol. fraction?

Brownian Hard Spheres

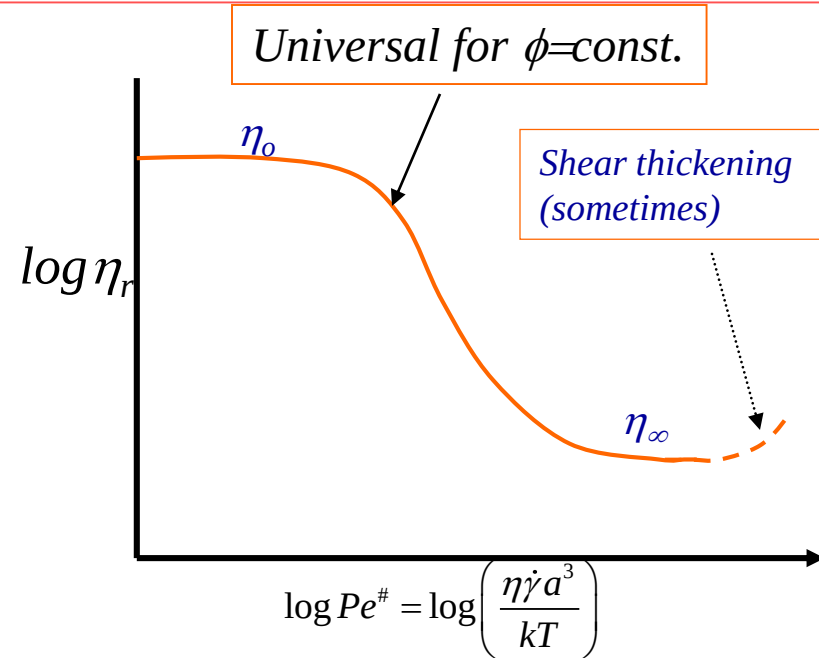
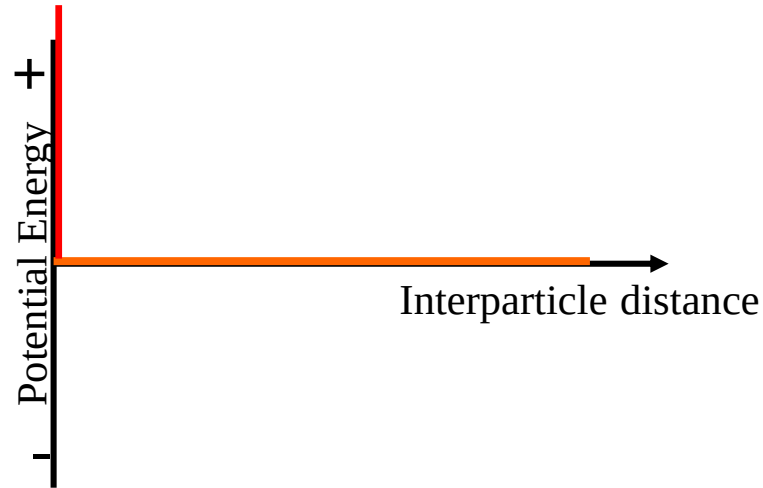
Let's look at the details of hard spheres

Brownian hard spheres

- ***Neglect dispersion, electrostatic, and steric interactions***
- ***Interparticle forces: Brownian, hydrodynamic***
- ***Resort to Peclet[#]***
$$Pe^{\#} = \frac{\eta_s \dot{\gamma} a^3}{kT}$$
- ***At higher particle concentration, more appropriate to replace η_s with η (suspension)***

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Brownian Hard Spheres



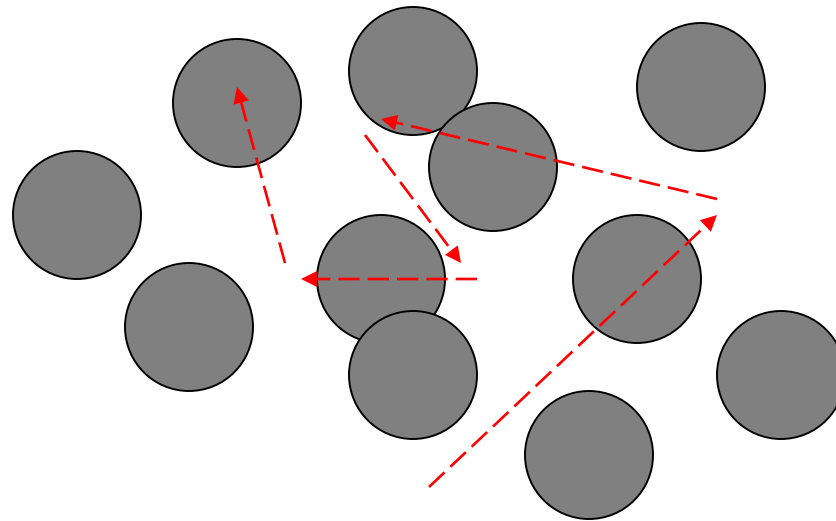
- *Universal behavior for monodisperse system*
Independent of τ , η_s , and particle size
- *Polydispersity can affect behavior*

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Non-dilute Hard Sphere Systems

Effect of Shear Deformation

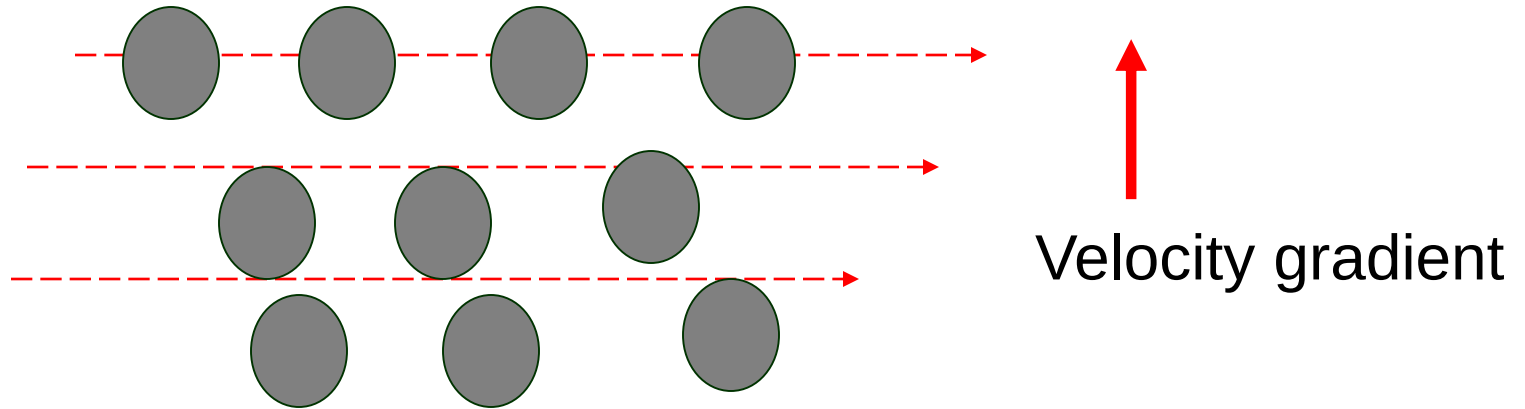
Low Rates of Deformation



**Randomising effect of thermal diffusion (“Brownian motion”)
leads to state of high relative viscosity**

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High Rates of Deformation

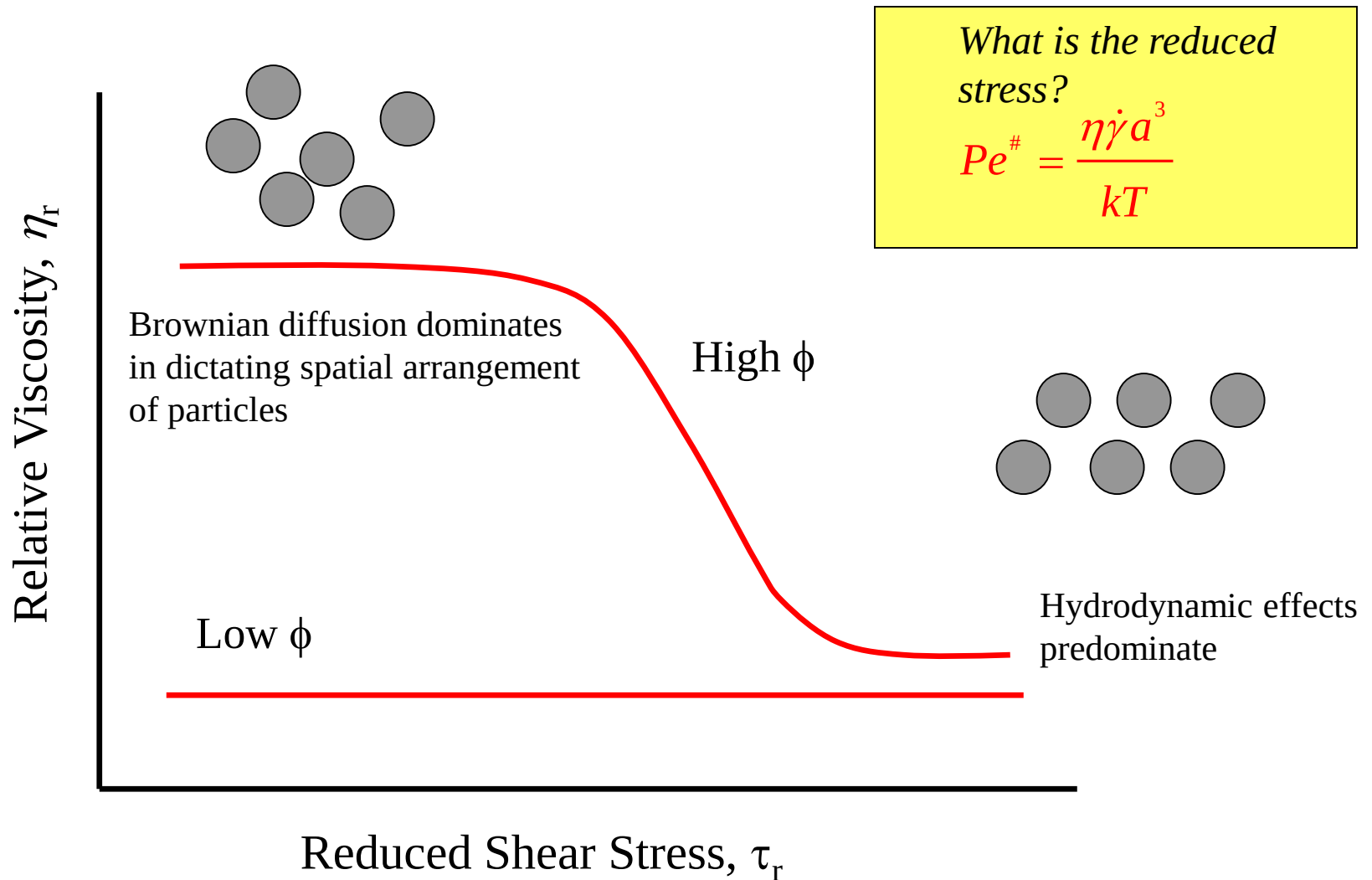


Predominance of hydrodynamic effects which arise as a consequence of the imposed shear deformation

Spatially, the particles adopt a lamellar arrangement which corresponds to a state of low relative viscosity

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Generalised Viscosity - Stress Plot



What is the reduced stress?

$$Pe^\# = \frac{\eta \dot{\gamma} a^3}{kT}$$

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Brownian Hard Spheres, cont'd

Of interest are:

- $\eta_o = f(\phi)$ ϕ = volume fraction of solid
- $\eta_\infty = f(\phi)$ ϕ = volume fraction of solid

Use Krieger-Dougherty relation:

$$\eta_{rx} = \left[1 - \frac{\phi}{\phi_{\max}} \right]^{-2.5\phi_{\max}}$$

relative

$$\begin{aligned} x=0 & \Rightarrow \eta_o \\ x=\infty & \Rightarrow \eta_\infty \end{aligned}$$

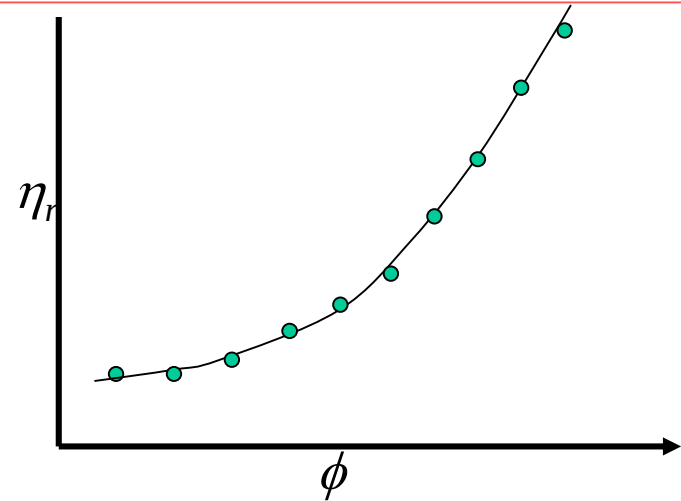
$\phi_{\max}$ = max packing fraction

$$\begin{aligned} \dot{\gamma} \rightarrow 0, & \quad \phi_{\max} = 0.64 - 0.63 \\ \dot{\gamma} \rightarrow \infty, & \quad \phi_{\max} = 0.71 \end{aligned}$$

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Krieger-Dougherty Relation

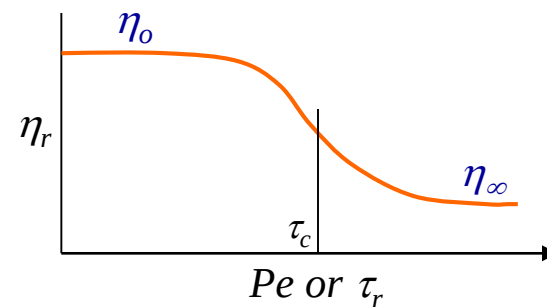
$$\eta_{\text{rx}}^{\text{relative}} = \left[1 - \frac{\phi}{\phi_{\text{max}}} \right]^{-2.5\phi_{\text{max}}}$$



Krieger also proposed:

$$\frac{\eta - \eta_{\infty}}{\eta_o - \eta_{\infty}} = \frac{1}{1 + \tau_r / \tau_c}$$

τ_c = reduced shear stress for which viscosity reaches average value between two Newtonian plateau

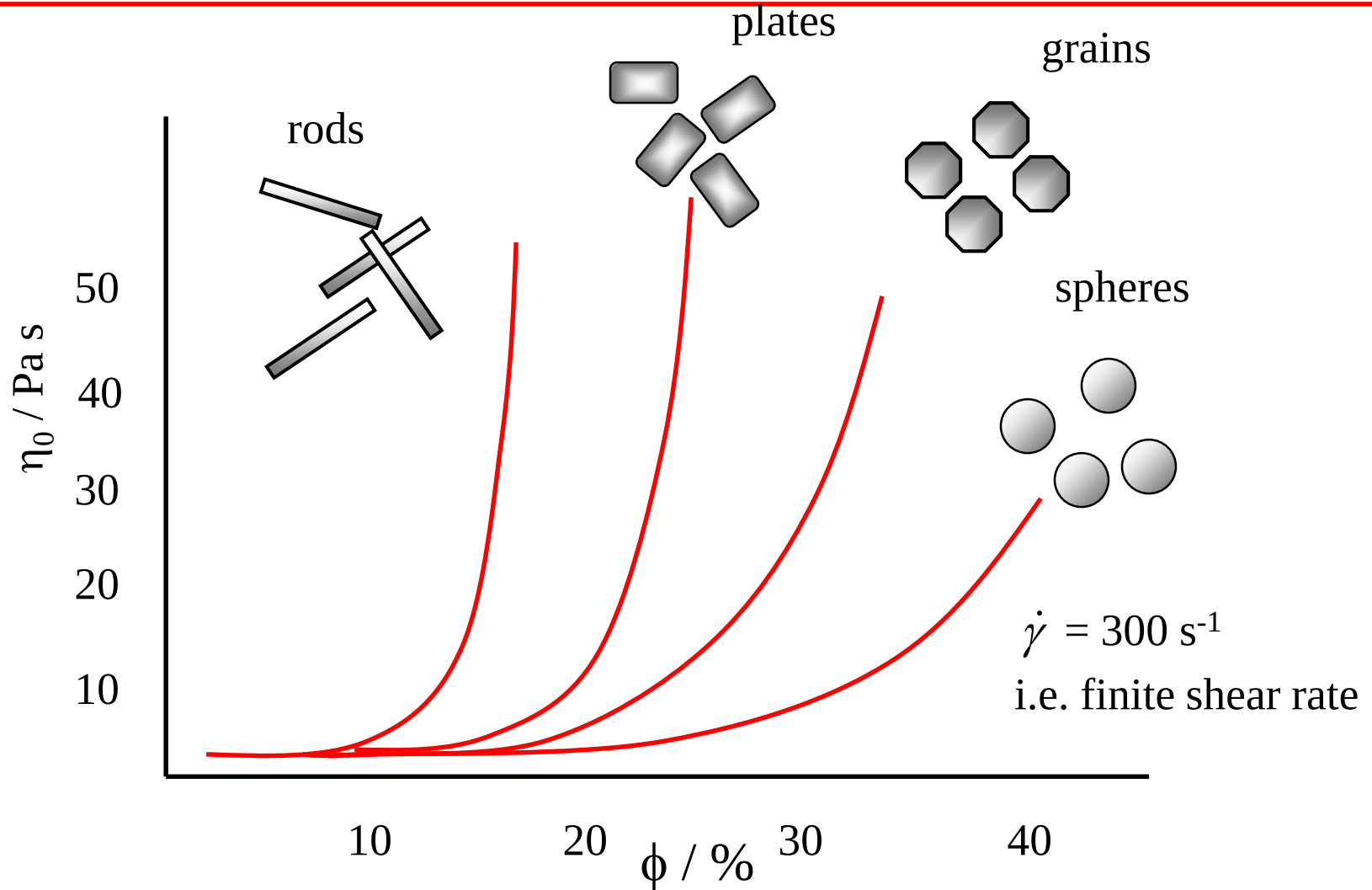


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Particle Geometry		$[\eta]$	$\phi_{\max}$
Spheres: Simple cubic		2.50	0.5236
Body-centred cubic		2.50	0.6800
	Random close-packed	2.50	0.6392
Random loose-packed		2.50	0.5886
Rods or ellipsoid, random packed:			
Aspect ratio = 2		2.58	0.68
Aspect ratio = 10		5.93	0.42
Fibres , random packed:			
Aspect ratio = 7		3.80	0.374
Aspect ratio = 21		6.00	0.233

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Influence of Particle Shape



$\phi_{\max}$ increases - rods > plates > grains > spheres

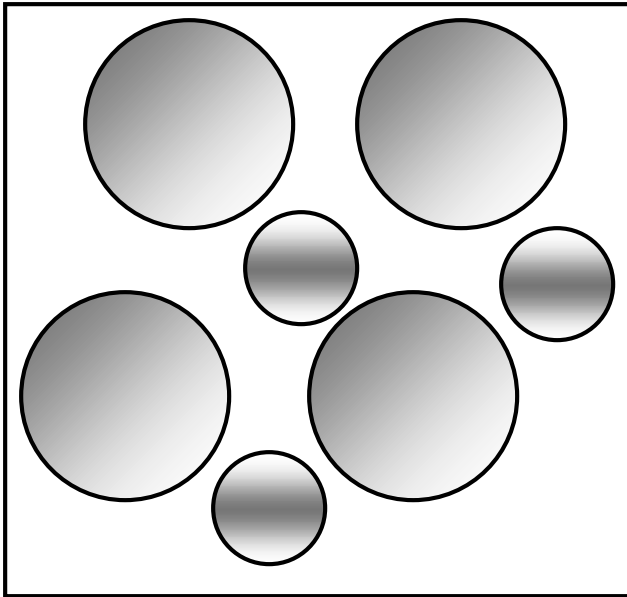
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Effect of Polydispersity / Particle Size Distribution

Several studies carried out on bimodal systems

Polydispersity increases ϕ_{max}

(smaller particles fit in interstitial spaces between larger particles)



Often employed in emulsion polymer technology to increase solids at a given viscosity

$$\frac{\eta}{\eta_s} = \underbrace{\left(1 - \frac{\phi_1}{\phi_{1max}}\right)^{-[\eta_1]\phi_{1max}}}_{\text{fine fraction}} \underbrace{\left(1 - \frac{\phi_2}{\phi_{2max}}\right)^{-[\eta_2]\phi_{2max}}}_{\text{coarse fraction}}$$

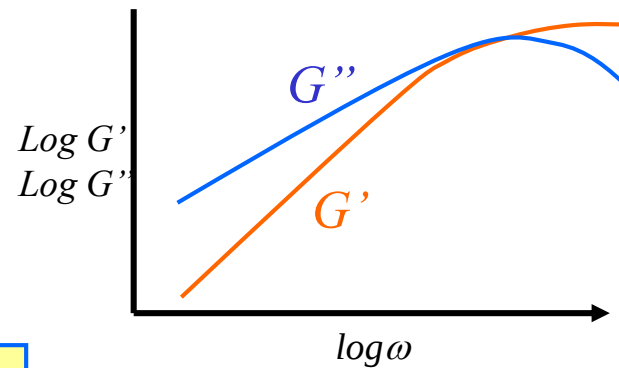
Systems with a complex, continuous distribution of particle sizes - impossible to predict ϕ_{max}

Maximum packing fraction must be determined experimentally

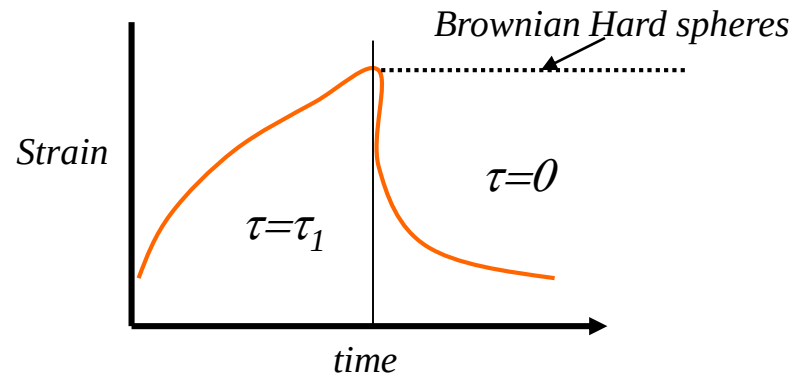
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Brownian Hard Spheres cont'd

G, G'' of Brownian hard spheres:



Creep and creep recovery:



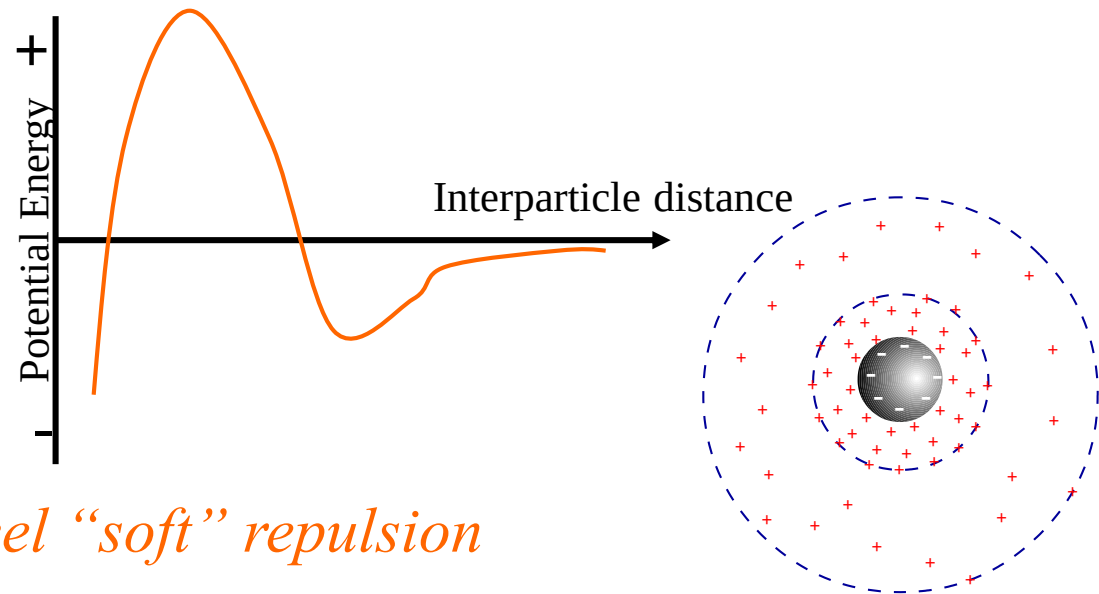
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Stable Suspensions

Sterically or electrostatically stabilized

Electrostatically stabilized system

Submicron fillers where we have colloidal interactions



Particles will feel “soft” repulsion

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Electrostatically Stabilized Suspensions

At low ϕ :

$$\eta = \eta_o \left(1 + \frac{5}{2} \phi + H_o \phi^2 \right)$$

H_o is a function of charges, etc.

Problem

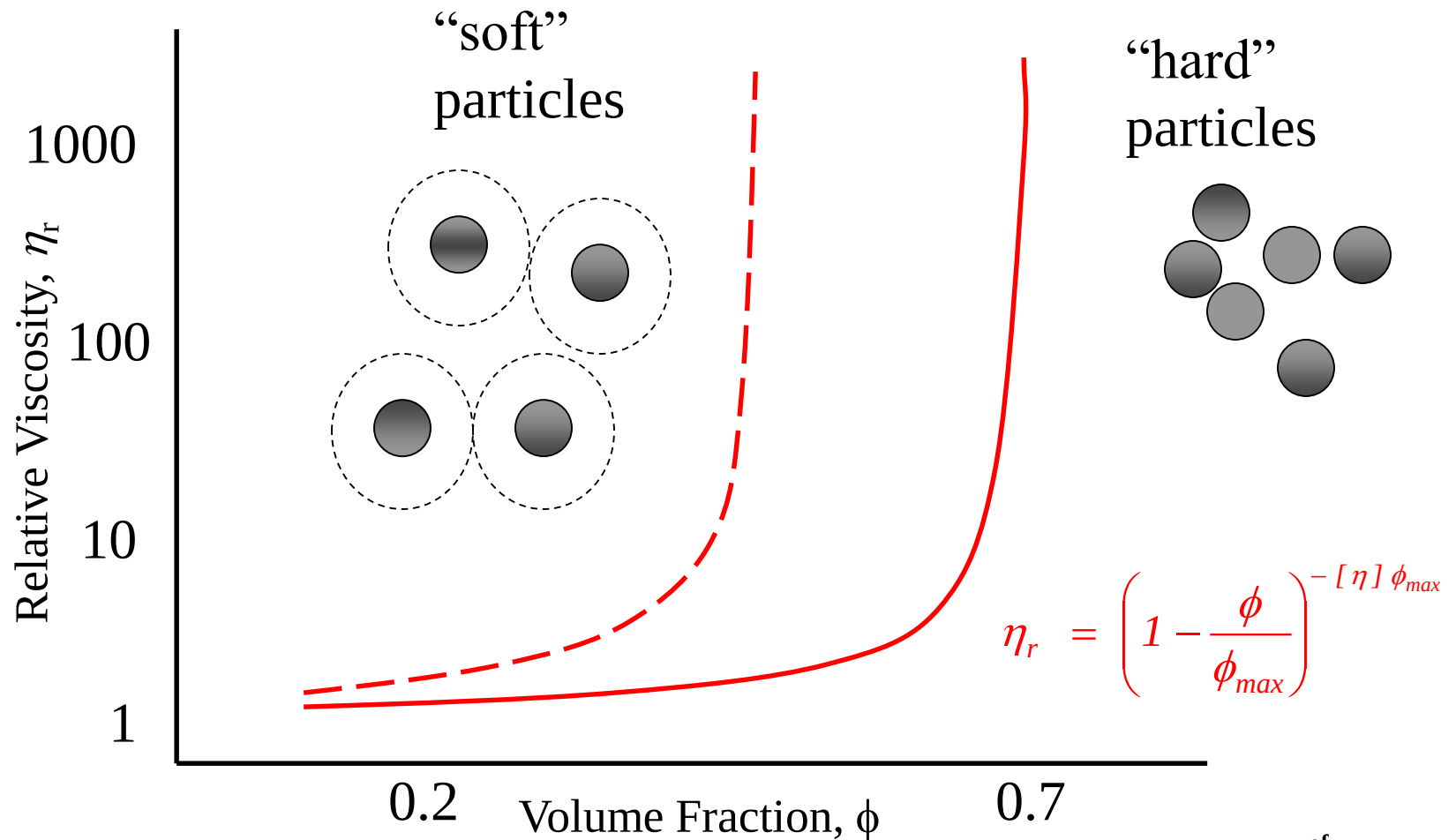
Have two systems at $\Phi=0.3$,

- electrostatically stabilized
- Brownian Hard sphere

Which will have a higher viscosity?

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Viscosity vs Particle Volume Fraction – Influence of Colloidal Forces

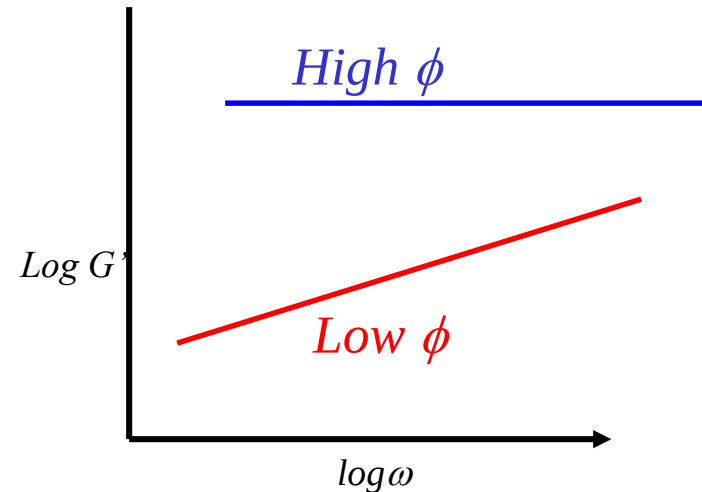


Electrostatically Stabilized Suspensions

Plot the dynamic frequency spectrum of G' for different ϕ

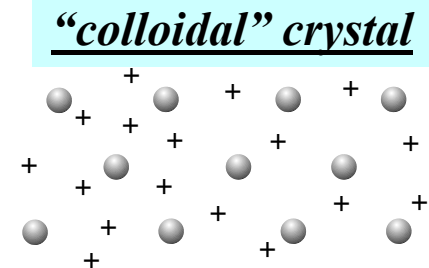
At low ϕ

G' frequency dependent and lower than G'' at low frequencies



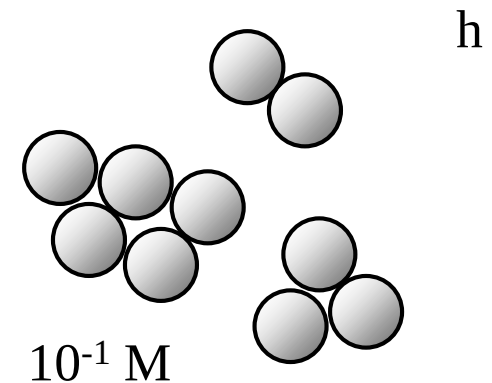
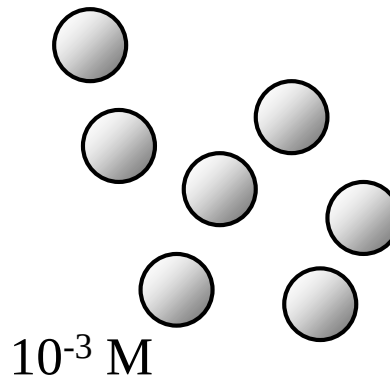
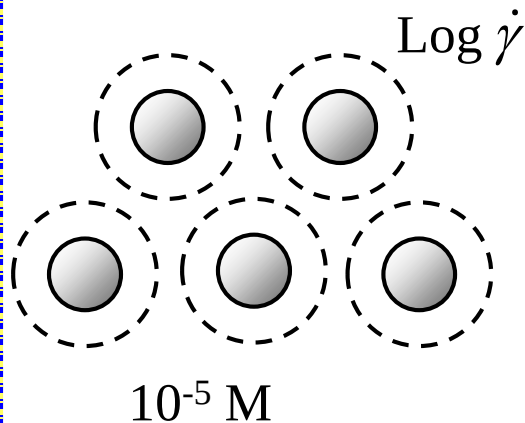
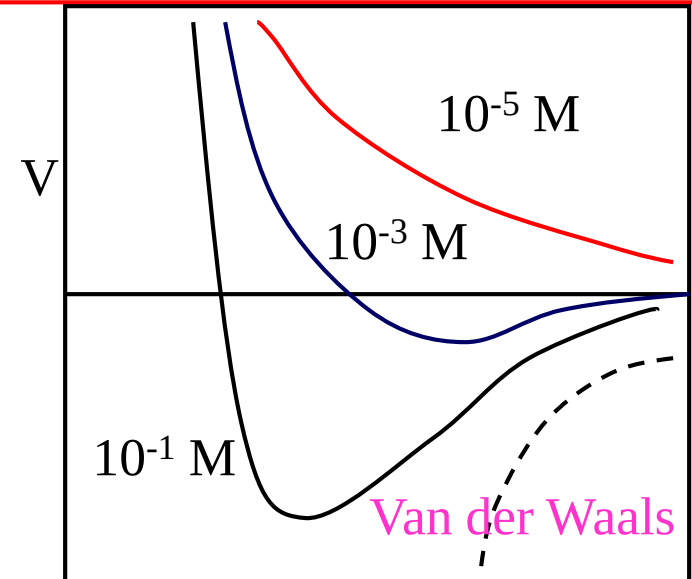
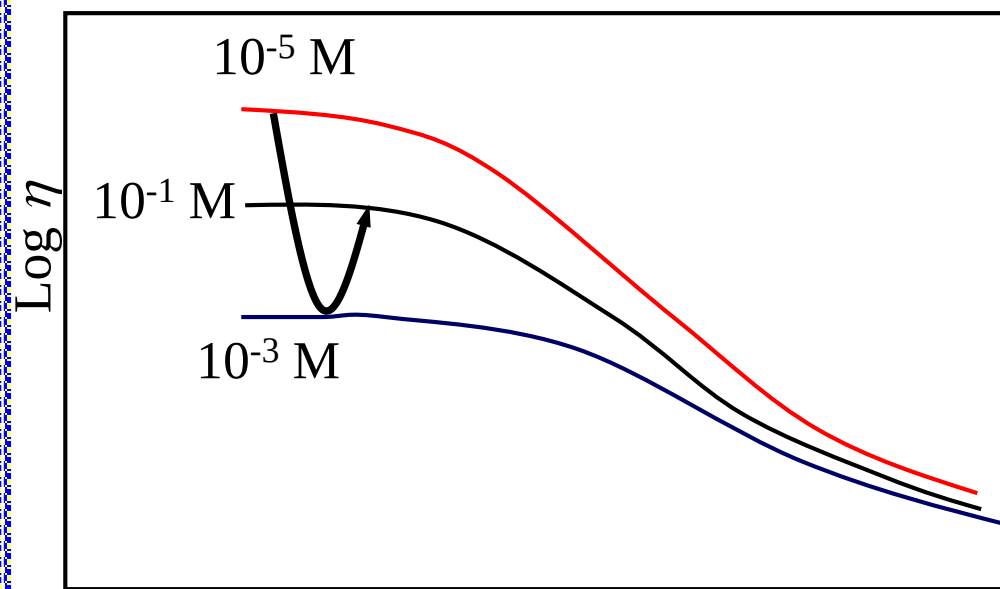
At high ϕ

Particles position themselves as far apart as possible in a lattice structure to minimize repulsion between them.



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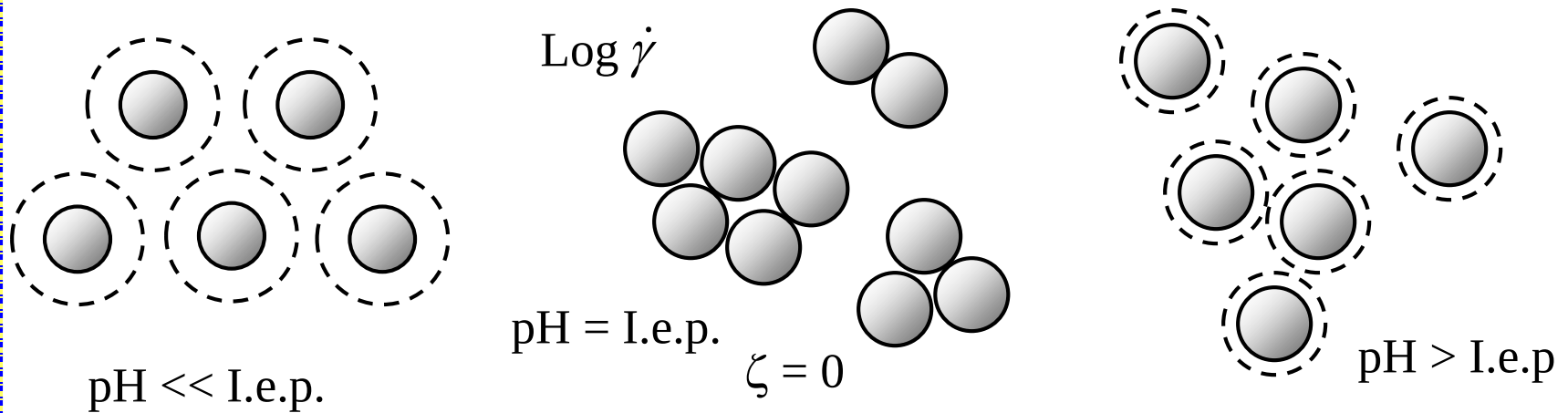
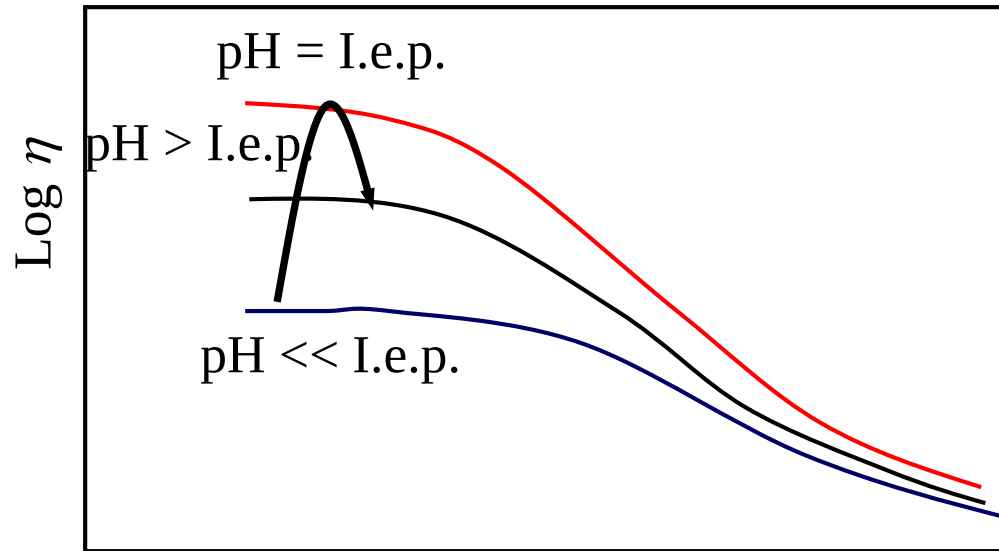
Particles with strong electrostatic repulsion: Electrolyte Effects



Increasing ionic strength →

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Particles with strong electrostatic repulsion: pH Effects



Increasing pH →

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